Lab Objectives

- LO 3.1 Calculate and interpret measures of location.
- LO 3.2 Calculate and interpret measures of dispersion, shape, and association.
- LO 3.3 Detecting Outliers.

Summary of Measures:

1- Measures of Location: mean – median – mode – percentile.
2- Measures of dispersion: range - The interquartile range (IQR) - The variance and the standard deviation - The coefficient of variation (CV).
3- Measures of Shape: Skewness - The kurtosis coefficient.
4- Measure of Association: The correlation coefficient.

Exercises

Question 1 (3.1 file: Prime)

You are a data analyst at Amazon, and your manager has shared data on the annual spending (in USD) of 100 Amazon Prime customers. The goal is to understand customer spending behavior to help design a new loyalty program targeting typical and high-value customers.

Task:
Use the dataset to summarize and interpret customer spending patterns.

a. Describe the typical annual spending behavior of Prime customers.
b. Identify spending thresholds that distinguish lower, average, and higher-spending customers, and interpret what these insights might mean for marketing decisions.

Answer 1

a. Understanding the Typical Spending Behavior:
   To describe typical customer spending, calculate both the mean and the median of annual expenditures.
   - Mean (Average Spending): $1,306.94
   - Median (Middle Spending Value): $1,287.50

   Interpretation:
   - The mean and median are quite close, suggesting that spending among Amazon Prime customers is fairly balanced, with no extreme outliers significantly skewing the data.
   - On average, a typical Prime customer spends around $1,300 per year.

b. Identifying Spending Segments (Quartiles)

Calculate the first quartile (Q1) and the third quartile (Q3) to understand how spending varies among customers.

- Q1 (25% spend less than): $1,118.75
- Q3 (75% spend less than): $1,574.25

Interpretation:

- About 25% of customers spend below $1,119, indicating lower-spending members.
- About 25% of customers spend above $1,574, representing higher-value or premium customers.
- The range between Q1 and Q3 shows where the majority (middle 50%) of customers fall, roughly between $1,100 and $1,600.
- This insight helps marketing teams, for example, target high-value customers for loyalty rewards and design offers to engage lower spenders.

Question 2 (3.1 file: Franchise)
You are part of the Business Analytics team for a national fast-food chain that operates 100 restaurants. The dataset provides information on Net Profit, Counter Sales, and Drive-Through Sales, all measured in millions of dollars. Management wants to better understand how performance varies across locations.

Task:
Use the dataset to explore and interpret key performance patterns.

a. Summarize the distribution of each variable to highlight the overall performance spread across restaurants.
b. Based on your results, explain what the typical net profit indicates about the general financial performance of the restaurants.
c. Interpret what the lower and upper quartiles suggest about differences in sales performance between counter and drive-through operations.

Answer 2

a. Summarizing Restaurant Performance
To explore overall performance across the 100 restaurants, calculate the mean, median, mode, standard deviation, and variance for each metric:

Mean	0.896	Mean	6.523	Mean	4.762
Median	0.8	Median	6.1	Median	4.35
Mode	0.6	Mode	8.4	Mode	7.7
Standard Deviation	0.430860075	Standard Deviation	2.01443553	Standard Deviation	2.282864584
Sample Variance	0.185640404	Sample Variance	4.057950505	Sample Variance	5.211470707

Interpretation:

- On average, each restaurant earns $0.9M in profit, with counter sales generating more revenue than drive-through sales.
- Drive-through sales show slightly higher variability (std. dev. = 2.28) compared to counter sales (2.01), meaning performance differs more widely between branches in the drive-through channel. This suggests inconsistent efficiency or location performance in drive-through operations.

b. Interpreting the typical Net Profits: Median Net Profit = $0.8M

Interpretation:
- Half of the restaurants earn less than $0.8M, while the other half earn more.
- This represents the "typical" restaurant's profit and provides a realistic benchmark for assessing whether a given branch is performing above or below average.

c. Interpreting Percentiles

- 25th Percentile (Counter Sales): $4.7M
  About 25% of restaurants generate less than $4.7M from counter sales, indicating lower-performing branches.

- 75th Percentile (Drive-Through Sales): $7.7M
  About 25% of restaurants generate more than $7.7M from drive-through sales, highlighting top-performing outlets in that channel.

- Overall Interpretation:
  The percentile results help management segment restaurant performance, identifying which branches fall into the bottom, middle, and top quartiles for targeted improvement or reward programs.

Question 3 (3.2 file: Spend)
You are working as a data analyst for a national statistics agency studying household spending behavior.
The dataset contains annual expenditures on food and travel for 500 households. Policymakers want to understand how spending patterns differ between these two categories.

Task:
Use the data to analyze and interpret the characteristics of each spending type.

a. Compare the overall spending levels and variability of food and travel expenditures. Discuss which category shows greater absolute and relative variation.
b. Examine the shape of each distribution and explain what the skewness values reveal about household spending tendencies.
c. Assess the peakedness or flatness of each distribution using kurtosis, and interpret what it suggests about how most households spend in each category.

Answer 3

a. Comparing Spending and Variability
Calculate the mean, standard deviation, and coefficient of variation (CV) for each spending category:

Category	Mean ($)	Std. Dev ($)	Coefficient of Variation (CV)
Food	4,420.28	1,741.30	0.394
Travel	2,405.27	1,137.40	0.473

Interpretation:

- On average, households spend more on food than on travel.
- Absolute variability (standard deviation) is higher for food, meaning the dollar differences are larger for food spending.
- Relative variability (CV) is higher for travel (0.473 vs. 0.394), indicating that spending on travel is less consistent across households relative to its average.

b. Examining Skewness (Shape of Distribution)

- Food skewness: 0.839 (moderately positively skewed)
- Travel skewness: 0.619 (moderately positively skewed)

Interpretation:

- Positive skewness indicates that most households spend below the average, with a few households spending significantly more.
- Spending on food is slightly more skewed than travel, meaning there are more extreme high-spending households in food compared to travel.

c. Assessing Peakedness (Kurtosis)

- Food kurtosis: 0.521: more peaked than normal
- Travel kurtosis: -0.301: flatter than normal

Interpretation:

- The distribution of food expenditures has a sharper peak, indicating that many households spend near the average, with few extreme values.
- Travel expenditures are flatter, showing a more even spread of spending amounts and fewer households clustered near the mean.

Question 4 (3.2 file: Test_Scores)
You are a data analyst supporting the Academic Affairs Office at a university. The office is reviewing data for 32 students, including their midterm and final exam scores, to assess grading consistency and overall student performance trends.

Task:
Use the dataset to carry out a comprehensive student performance analysis.

a. Compare the average performance and variability between the midterm and final exams.

b. Discuss which exam shows greater absolute and relative variability in scores. What might this indicate about the exams or students' performance consistency?
c. Examine the relationship between midterm and final results. What does the correlation tell you about how performance in one exam relates to the other?

Answer 4
a. Comparing Average Performance and Variability

- Calculate the mean and standard deviation for both exams:

Exam	Mean	Standard Deviation
Midterm	80.75	14.36
Final	82.28	17.63

Interpretation:

- The final exam average is slightly higher than the midterm, suggesting an overall improvement or higher scoring in the final.
- The final exam also has greater variability (std. dev. = 17.63 vs. 14.36), indicating that student performance was more spread out, with some students doing much better or worse than the average.

b. Exam Variability Comparison

- Both absolute (standard deviation) and relative (coefficient of variation) measures show that the final exam scores are more dispersed than midterm scores.
- This could indicate that the final exam was more challenging or allowed for a wider range of performance differences among students.

c. Relationship Between Midterm and Final Performance

- Correlation coefficient: 0.552

Interpretation:

- There is a moderate positive correlation between midterm and final scores, meaning students who performed well in the midterm generally performed well in the final.
- This suggests some consistency in student performance.

Question 5 (3.34 "File: Tech_Energy").
You are part of an investment analytics team evaluating two mutual funds — one focused on Technology and the other on Energy — using 37 years of annual return data (in %).
Your goal is to assess the performance patterns and identify any unusual return behavior that could affect long-term analysis.

Task:
Use the dataset to explore and interpret anomalies in both investment categories.

a. Visualize the annual returns for the Technology fund using a boxplot and comment on whether potential outliers appear.

b. Verify your findings using z-scores. Are the results consistent with the boxplot? Explain any differences.
c. Repeat the same analysis for the Energy fund, identifying and confirming potential outliers.
d. Based on your findings, replace any identified outliers with missing values (NAs) and determine the new average annual returns for both funds. Discuss how these changes might influence investment decisions.

Answer 5

a. Technology Fund – Boxplot Analysis

- Visual inspection of the boxplot shows one or two points above the upper whisker, suggesting potential high-return outliers.
- The central 50% of returns is concentrated between the lower and upper quartiles, with a few extreme positive returns pulling the mean upward.

b. Technology Fund – Z-Score Analysis

- Minimum z-score: -1.88
- Maximum z-score: 3.12

Interpretation:

- Using a typical cutoff of $|z| > 3$ to flag outliers, the maximum z-score (3.12) indicates one high-return outlier.
- This is consistent with the boxplot, confirming the presence of an extreme positive return.

c. Energy Fund – Boxplot Analysis

- The boxplot shows one low-return point below the lower whisker, suggesting a potential negative outlier.
- The majority of returns are tightly clustered around the median, indicating less extreme variation compared to Technology.

Energy Fund – Z-Score Analysis

- Minimum z-score: -2.79
- Maximum z-score: 1.83

Interpretation:

- No z-scores exceed ±3, so technically there are no extreme outliers by this criterion.
- The boxplot indicates a possible mild outlier, but z-scores suggest it is not extreme enough to be statistically flagged.
- Differences arise because boxplots flag mild outliers beyond 1.5×IQR, whereas z-scores flag extreme deviations from the mean.

d. Handling Outliers and Recalculating Averages

- After identifying outliers via the boxplots, replace them with NAs and recalculate the averages.
- Removing outliers generally reduces the mean for Technology slightly, and may have a smaller effect on Energy, due to the fewer extreme returns.

Interpretation:

- Adjusted averages provide a more representative estimate of typical annual returns.
- This approach helps analysts avoid overestimating/underestimating fund performance due to extreme observations, supporting more accurate long-term investment planning.